

	Owner(s): Quality Manager	Work Instruction No.:	
	Approval(s):	Effective Date:	1/17/2026
		Rev. No.: 2	Controlled
		Page: 1 of 4	
Title:		Laboratory Scope & Manual	

PURPOSE:

To outline the scope of the laboratory at the Company Name Mason facility as defined in the Laboratory Scope and manual.

SCOPE:

This work instruction applies to the Company Name Mason facility.

REVISION:

Update reference documents
Updated Lab scope (1/17/2026)

RESPONSIBILITY:

1. The Quality Manager is responsible for the implementation, maintenance, and approval of this work instruction. Additional responsibilities may be delegated to properly trained and/or authorized personnel.

DEFINITIONS:

1. None.

REFERENCE DOCUMENTS:

1. PR-42-002, Control of Records
2. WI-76-901, Calibration
3. WI-76-003, Microstructure Preparation
4. WI-76-004, Mixture of Etching Solution
5. WI-76-005, Tensile Testing
6. WI-76-006, Pickle Testing
7. AIAG Statistical Process Control Reference Manual

WORK INSTRUCTION:

1. The Quality Lab has the responsibility for gage inspection including calibration and Gage R&R. Other responsibilities include PPAP verification, microstructure analysis and test fixtures.
2. Personnel performing testing and/or calibration are qualified by benefit of previous work experience, formal training, certification or demonstration of competencies.
3. Product submitted for testing is adequately identified to preclude unintended use and is traceable for the life of the testing activity.
4. Environmental controls are established and recorded by maintenance to maintain the integrity of the testing quality results. Reference work instruction **WI-76-901, Calibration**.
5. Testing is performed in accordance with documented testing instructions and/or customer requirements.
6. Methods for SPC are outlined in the AIAG SPC Manual.
7. Calibration of inspection, measuring and test equipment is performed per work instruction **WI-76-901, Calibration**.

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WORK INSTRUCTION:

8. The Laboratory Scope (Attachment A) outlines the inspection, test, and calibration activities that the laboratory is qualified to perform.
9. Any external laboratories used for inspection and/or testing meet ISO/IEC 17025 or national equivalent. Certification is maintained by the Quality Department.
10. Records of laboratory tests, inspections, and calibrations are retained in accordance with procedure **PR-42-002, Control of Records**.

ATTACHMENT A:

LABORATORY SCOPE

Inspection / Test / Calibration Performed	Equipment Used	Method or Standard Used
Measurement of parts and control devices	Pro-T Advance Coordinate Measuring Machine (3 dimensional) X = 5000mm Y = 2000mm Z = 2200mm	CMM-OS Caligo Software Customer Specific Requirements
Measurement of parts and control devices	Accura Coordinate Measuring Machine (3 dimensional) X = 1600mm Y = 3000mm Z = 1400mm	HOLOS-NT/UX software Calypso Software Customer Specific Requirements
Measurement of parts and control devices	Pro-T Select Coordinate Measuring Machine (3 dimensional) X = 3200mm Y = 1600mm Z = 2100mm	CMM-OS Caligo Software Customer Specific Requirements
Rockwell Hardness Testing (B,C)	Tinius Olsen/Newage Rockwell Tester	Customer Specific Requirements or ASTM E-18
Micro Hardness Testing (HV)	Leco LM 100AT Micro Hardness Tester	Customer Specific Requirements or ASTM E-386
Nital Acid Prep	Nitric Acid, Denatured Alcohol Bath, Grad. Cyl, Beaker, Pipette, Storage Bottle	WI-76-003, Microstructure Preparation WI-76-004, Mixture of Etching Solution Customer Specific Requirements

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Inspection / Test / Calibration Performed	Equipment Used	Method or Standard Used
		ASM Metallography & Microstructure Handbook
Tensile Testing	Zwick 100KN Tensile Tester	WI-76-005, Tensile Testing Customer Specific Requirements ASTM E8
Metallographic Sample Prep	Leco PR-36 Mount Press/ Leco GPX 300 Grinder/ Polisher	WI-76-003 Microstructure Preparation ASTM E3
Metallographic Evaluation	Olympus GX 51 Metallography	Customer Specific Requirements ASTM E883
Weld Analysis	Calipers Nut push-out Tester Torque Wrench Tessonics UT Test Unit	Tessonics Training Course Manual, ISO 10447 Resistance Welding, ASTM E164 STD Practice for Contact Ultrasonic Testing of Weldments, Customer Specific Requirements, Internal Work Instructions
Pickle Test	Leco MSX Saw/Etch Bath	MA-WI-76-006, Pickle Testing
Depth of Decarburization	Leco PR-36 Mount Press/ Leco GPX 300 Grinder/ Polisher/ Olympus SZX7/Olympus GX51 Metallograph/Pax-It Software	Customer Specific Requirements, ASTM E1077
Surface Roughness	Mitutoyo SJ-201P SR Tester (Rz/Rx)	Customer Specific Requirements SAE J911
Calibration of Checking Fixtures	CMM	CMM program
Calibration of Micrometers	Gage Blocks	Calibration instruction located in ProGage Calibration System
Calibration of Calipers	Gage Blocks	Calibration instruction located in ProGage Calibration System
Calibration of Height Gages	CMM	Calibration instruction located in ProGage Calibration System
Calibration of Dial Indicators	Gage Blocks	Calibration instruction located in ProGage Calibration System
Calibration of Pin Gages	Micrometer	Calibration instruction located in ProGage Calibration System
Calibration of Go/No Go Gages	Micrometer	Calibration instruction located in ProGage Calibration System
Calibration of Depth Micrometer	Gage Blocks	Calibration instruction located in ProGage Calibration System
Calibration of Snap Micrometer	Gage Blocks	Calibration instruction located in ProGage Calibration System

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Inspection / Test / Calibration Performed	Equipment Used	Method or Standard Used
Calibration of Radius Gage	Certified Radius Gage	Calibration instruction located in ProGage Calibration System
Calibration of Micro Indentation hardness tester	Certified Test Block	Calibration instruction located in ProGage Calibration System
Calibration of Masters	CMM	Calibration instruction located in ProGage Calibration System
Calibration of Inside Micrometers	Setting Rings	Calibration instruction located in ProGage Calibration System
Calibration of Gage Block	Gage Block Set, Micrometer	Calibration instruction located in ProGage Calibration System
Verification of Fixture	CMM results, Primary Master	Calibration instruction located in ProGage Calibration System